

23. (2 points)

A large insurance company has developed an Enterprise Risk Management (ERM) model which has the following characteristics:

- i. The ERM model will be maintained by internal employees across various functions (including actuarial, underwriting and planning) who will work on the model in addition to their regular duties.
- ii. The insurance company is exposed to a significant amount of earthquake risk. To quantify its earthquake risk, the company uses a single earthquake model from a third-party vendor.
- iii. The ERM model incorporates correlation among the insurance company's different lines of business using parameters developed individually by each line of business leader.
- iv. The insurance company reviews the simulation results from the ERM model and sets its capital requirements based solely on the amount required to achieve a sufficiently small probability of default.

Describe one weakness of each characteristic above and propose a possible improvement to address each weakness.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 23	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE(S): C1, C5, C6
SAMPLE ANSWERS	
<i>Sample responses are separated into the four main characteristics listed in the question:</i>	
Characteristic i • Work and scheduling conflicts with employees' regular jobs. Have some dedicated employees to work solely on ERM. • This introduces risk of delay of ERM implementation as employees may say they are too busy to do their ERM work on time. Have dedicated full time employees do ERM work. Characteristic ii • Results from cat models can vary greatly from version to version. Using multiple models is superior. • Only one model used. Should use multiple models to reduce internal systemic risk. Characteristic iii • Model should be parameterized by people with deep knowledge of subject. Correlation requires deep knowledge of multiples lines, not just one. Have ERM team consider correlations themselves in addition when selecting. • Parameters are built using different types of assumptions. Roundtable with actuary and LOB leaders to arrive at a consensus on parameters and parameter variance. • Each leader does not have enough of the overall firm knowledge to sufficiently estimate parameters. Should be owned at the corporate level. • Because of its high political sensitivity, individual business leaders may skew the parameters in their favor. Correlation assumptions should be owned by the CRO or ERM department. Characteristic iv • Significant impairment to operations could happen well before default. Consider metrics such as inability to service renewals, or ratings downgrades as thresholds. • Default risk too far out in the tail, tail is most uncertain and unreliable. Use different threshold. Capital necessary to maintain rating level. • Weakness: Capital set only to prevent default mostly protects policyholders – shareholders may have different risk tolerance. Improvement: Look at range of requirements necessary to not only recover but thrive from a CAT.	
EXAMINER'S REPORT	
Candidates were expected to know concepts of ERM and be able to identify weaknesses and know possible improvements. Candidates were expected to either explain why their weakness was a weakness or explain why their recommendation was an improvement to the weakness. Characteristic i • Candidates were expected to recognize that part time employees cannot dedicate time to the ERM process and that full time/dedicated employees were necessary.	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Characteristic ii

- Candidates were expected to know that there is uncertainty in the catastrophe model such as data quality and model/parameter risk and that using multiple models would address this issue. Candidates were expected to state why having one model was a weakness or what improvement multiple models would bring.

Characteristic iii

- Candidates were expected to know that individual Line of Business leaders are inexperienced or biased in setting correlations, and that assumptions be held by the CEO/CUO/CRO or in collaboration among them.

Characteristic iv

- Candidates were expected to provide a weakness that emphasizes the uncertain nature of modeling in the tail, protecting policyholders but not shareholders, or that losses lower than default can have negative impacts on the firm.

Common mistakes included:

- Not providing a weakness, or not explaining why the weakness identified was a weakness.
- Not explaining why the proposed solution was an improvement.